

No. of Functional Features Included in the Solution

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	ML Crop Recommendation	Predicts the optimal crop to cultivate based on NPK, pH, and climate inputs.	predict() in app.py / model.pkl	Done	35%
2	Crop Suitability Assessor	Calculates parameters alignment against ideal crop statistics ranges and outputs compatibility score.	suitability() in app.py	Done	25%
3	Agricultural Dashboard	Visualizes crop requirements (macronutrients, weather, pH) on comparison charts.	research.html / Chart.js	Done	20%
4	Parameter Range Validation	Validates inputs (non-negative limits, pH bounds, zero nutrient check) to prevent erroneous runs.	recommend() & suitability() in app.py	Done	10%
5	Static Cache Stats Engine	Caches averages, mins, and maxes for 22 crops to enable instant, local validations without lag.	crop_stats.json	Done	10%
6					
7					
8					

Feature Summary:

Metric	Count / Value
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Total Features Planned	5
Total Features Implemented	5
Core / Must-Have Features	3
Additional / Nice-to-Have Features	2 (Suitability gauges & comparative dashboard charts)
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Glassmorphic dashboard pages, slider inputs, compatibility gauges, and Chart.js graphs.	recommend.html, suitability.html, research.html
2	Backend / Logic	Flask server routing, Random Forest prediction engine, and soil threshold alignment logic.	predict(), suitability() in app.py
3	Database / Storage	Static file storage caching crop summaries and raw CSV datasets.	crop_stats.json, Crop_recommendation.csv
4	API / Integration	Stats API endpoint exposing cached crop limits for client-side Chart.js rendering.	/api/stats endpoint
5	Security / Authentication	Server-side type validations, negative number blocks, and form bounds verification.	Parameter sanitization checks